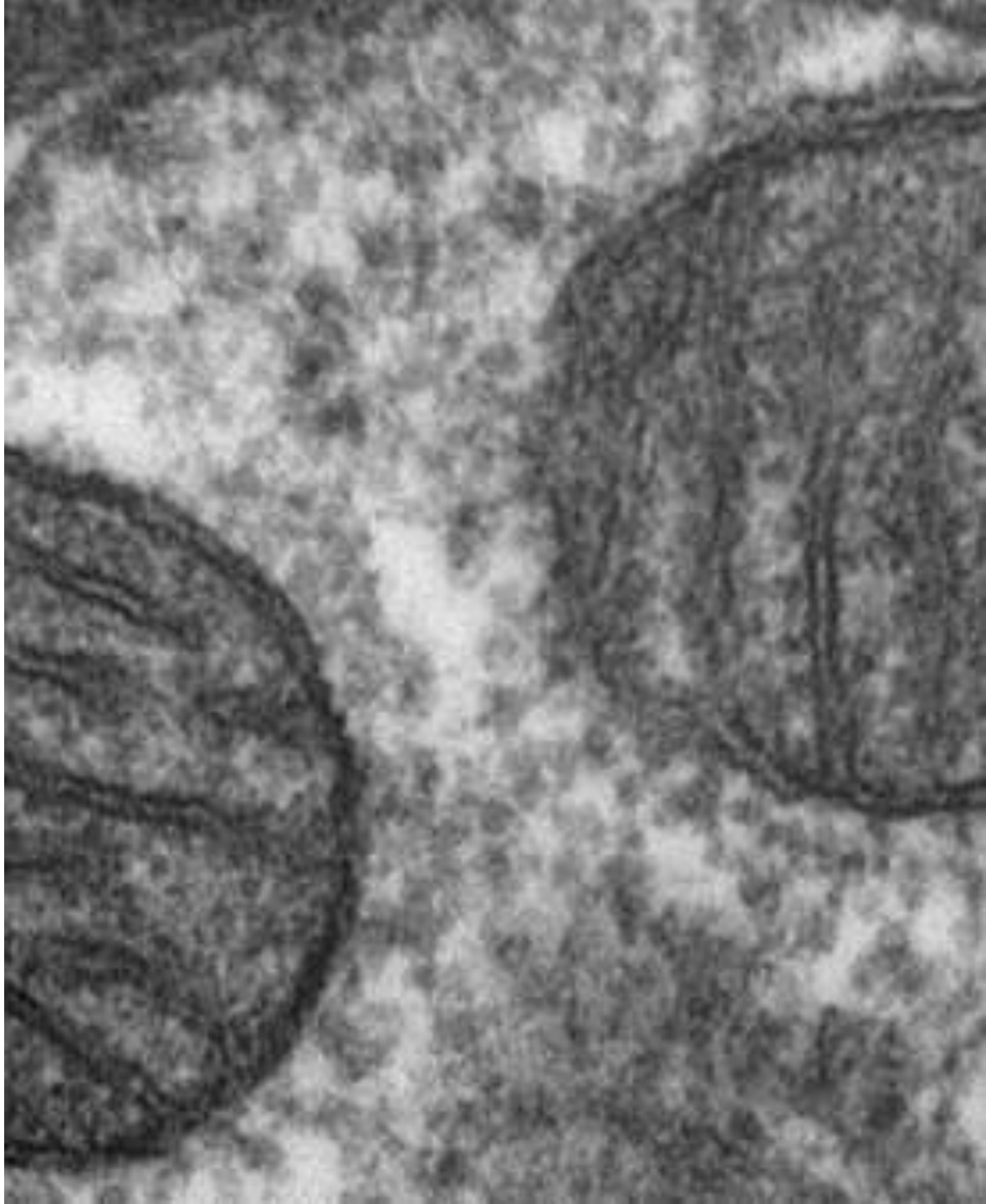
DP Biology – C1.2

Cell respiration

- What do hydrogen and oxygen contribute to releasing energy in cells?
- How is energy shared out and used within cells?

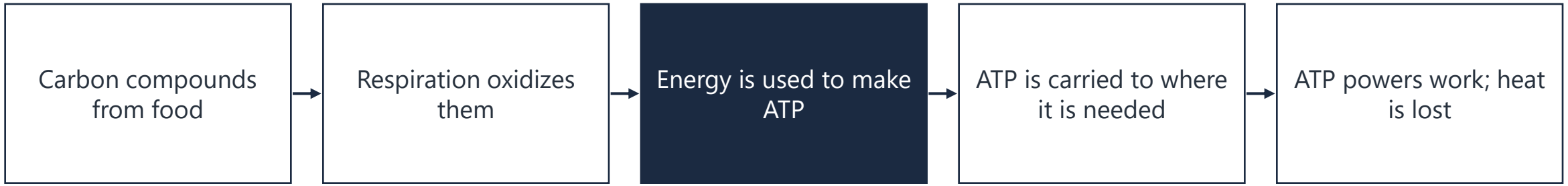
SL statements C1.2.1–6. Textbook pp. 405–411.
Image: mitochondria in a mammalian lung cell, electron micrograph. Louisa Howard, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	C1.2.1–3	Explain why ATP suits its role, what it is used for, and the energy changes between ATP and ADP
2	C1.2.4	Explain respiration as the system that makes ATP, and distinguish it from gas exchange
3	C1.2.5	Compare anaerobic and aerobic respiration in humans, with word equations
4	C1.2.6	Explain how a respirometer works, and calculate rates of respiration from data

The big picture: from food to work



Every cell needs a steady supply of energy for its work. That energy comes from carbon compounds such as glucose and fatty acids, but cells cannot use it directly.

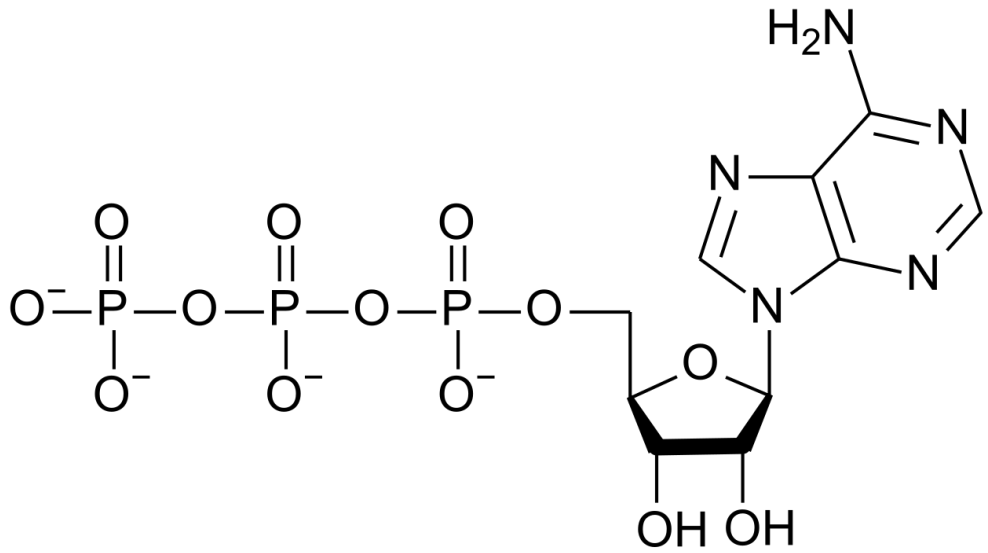
Cell respiration releases the energy step by step and uses it to make **ATP**. ATP then passes the energy on to whatever job needs doing: pumping ions, building proteins, contracting muscle.

Big question for this chapter

Why do you keep breathing hard for several minutes after a sprint, even though you have stopped running?

ATP: the energy currency of the cell

C1.2.1



Adenosine triphosphate. Mysid, public domain.

ATP (adenosine triphosphate) is a nucleotide: adenine, the sugar ribose and **three phosphate groups**.

It is called the **energy currency** of the cell: energy released in respiration is "paid in" to make ATP, and ATP is "spent" wherever energy is needed.

Cells keep only a small amount of ATP at any time and recycle it constantly.

Link

A1.2: ATP is built like an RNA nucleotide, with two extra phosphates.

Why ATP suits its job

C1.2.1

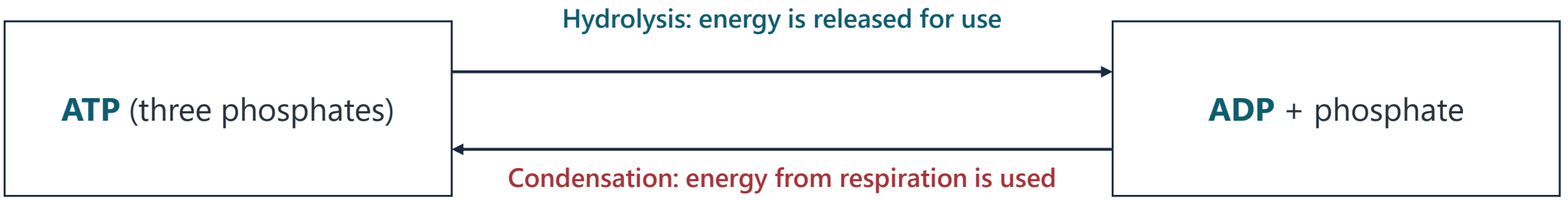
Property of ATP	Why it helps
Soluble in water	It moves easily through the cytoplasm to wherever it is needed
Cannot pass freely through membranes	It stays inside the cell or organelle where it was made
Releases a useful amount of energy	Removing one phosphate releases enough for most jobs, without wasting much
Easily rebuilt	ADP and phosphate can be turned back into ATP again and again

Key idea

ATP is not a store of energy but a carrier: it is made and used within seconds.

Energy transfers between ATP and ADP

C1.2.3



When ATP is **hydrolysed** to **ADP** (adenosine diphosphate) and phosphate, energy is released and can be used for work in the cell.

To rebuild ATP, phosphate is added back to ADP by **condensation**. This needs energy, which comes from cell respiration (or, in plants, from light).

Link

Hydrolysis and condensation are the same two reactions you met for polymers in B1.1: water is used up in one and released in the other.

What ATP is used for

C1.2.2

Process	Examples
Active transport	Sodium-potassium pumps in neurons; absorbing mineral ions into root hair cells
Synthesis of macromolecules (anabolism)	Making proteins, DNA, glycogen and starch
Movement of the whole cell	Beating of a sperm tail; amoeboid movement of white blood cells
Movement inside the cell	Muscle contraction; moving chromosomes in mitosis; vesicles moving along the cytoskeleton

Common mistake

ATP does not make energy. It passes energy on from respiration to these processes.

How much ATP does a body use?

An adult human contains only about **250 g** of ATP at any moment, enough for a minute or so of activity.

Yet each day the body hydrolyses and rebuilds roughly its **own body mass** of ATP, around 40 to 70 kg. Each ATP molecule is recycled hundreds of times a day.

During hard exercise, muscles can turn over ATP many times faster still.

Think about it

If ATP is used so fast, explain why a cell that cannot respire, for example because a poison has blocked its mitochondria, dies within minutes.

Respiration is not breathing

C1.2.4

	Cell respiration	Gas exchange
What it is	Chemical reactions that release energy from carbon compounds to make ATP	Oxygen enters and carbon dioxide leaves the organism
Where	Inside the cells of every living organism	Across an exchange surface such as alveoli, gills or leaves
Involves	Enzymes, glucose or other substrates, ATP	Diffusion of gases
Link	Uses oxygen and makes carbon dioxide	Supplies the oxygen and removes the carbon dioxide

Common mistake

Breathing (ventilation) helps gas exchange, which supplies respiration. They are three different things.

Respiration: a system for making ATP

C1.2.4

Cell respiration uses the energy released from carbon compounds to make ATP. **Glucose** and **fatty acids** are the main substrates, but almost any carbon compound can be respired if it is broken down first, including amino acids.

Substrate	Energy per gram	When it is used
Carbohydrate (glucose, glycogen)	About 17 kJ	First choice for most cells; the brain relies on it
Lipid (fatty acids)	About 37 kJ	At rest and in long exercise; only aerobic respiration
Protein (amino acids)	About 17 kJ	Mainly in starvation, after other stores run low

Link
B1.1.11: lipids store about twice as much energy per gram as carbohydrate.

Check-in: ATP and respiration

C1.2.1–4

1. State three properties of ATP that suit it to its role.
2. List three processes in cells that use ATP.
3. Describe what happens when ATP is hydrolysed.
4. Distinguish between cell respiration and gas exchange.
5. State two substrates for cell respiration.

How to use this

Answer without your notes, then compare with a partner.

Anaerobic respiration in humans

C1.2.5

glucose → lactate + a small yield of ATP

Anaerobic respiration needs **no oxygen**. It happens in the **cytoplasm** and uses only **glucose**.

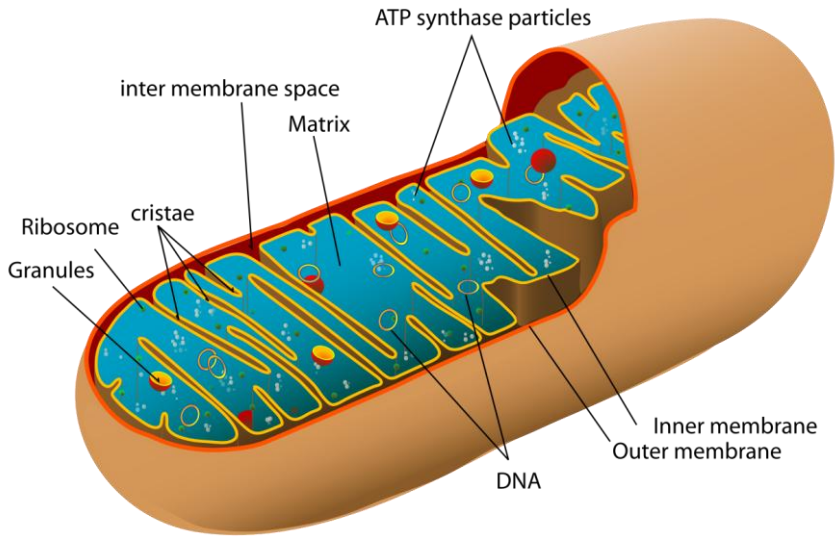
Each glucose gives only a **small** amount of ATP, but the ATP is made **very quickly**. In humans the waste product is **lactate**, which builds up in muscles during intense exercise.

When is it useful?

During a short burst of maximum effort, such as a 100 m sprint or lifting a heavy weight, muscles need ATP faster than oxygen can be delivered. Anaerobic respiration fills the gap for up to a couple of minutes.

Aerobic respiration in humans

glucose + oxygen → carbon dioxide + water + a large yield of ATP



A mitochondrion. LadyofHats, public domain.

Aerobic respiration starts in the cytoplasm and continues in the **mitochondria**.

It needs **oxygen**, and it can use glucose, **fatty acids** and other substrates.

The carbon compounds are fully **oxidized** to carbon dioxide and water, giving a **large yield** of ATP, about 15 times more per glucose than anaerobic respiration.

Guiding question

Hydrogen is removed from the substrate and finally combines with oxygen to form water. That is what oxygen contributes.

Comparing anaerobic and aerobic respiration

C1.2.5

	Anaerobic	Aerobic
Oxygen needed?	No	Yes
Substrates	Glucose only	Glucose, fatty acids and others
Where in the cell	Cytoplasm only	Cytoplasm, then mitochondria
Products in humans	Lactate	Carbon dioxide and water
Yield of ATP per glucose	Small	Large, many times greater
Speed of ATP production	Fast	Slower

Which type does the body use, and when?

C1.2.5

Activity	Main source of ATP	Why
Resting, walking	Aerobic	Plenty of oxygen reaches the muscles
100 m sprint (about 10 s)	Anaerobic (and stored creatine phosphate)	ATP needed faster than oxygen can be supplied
800 m race (about 2 min)	Both, roughly equally	Intense, but long enough for aerobic to contribute
Marathon (2 to 4 h)	Almost all aerobic	Steady pace; fatty acids and glycogen used

Try it

Explain why marathon runners can use fatty acids as fuel but sprinters cannot.

After the race: repaying the oxygen debt

C1.2.5

After intense exercise, breathing and heart rate stay high for several minutes. The extra oxygen taken in is used to:

break down **lactate** (mostly in the liver), converting much of it back to glucose;

rebuild stores of **ATP** and creatine phosphate in muscles;

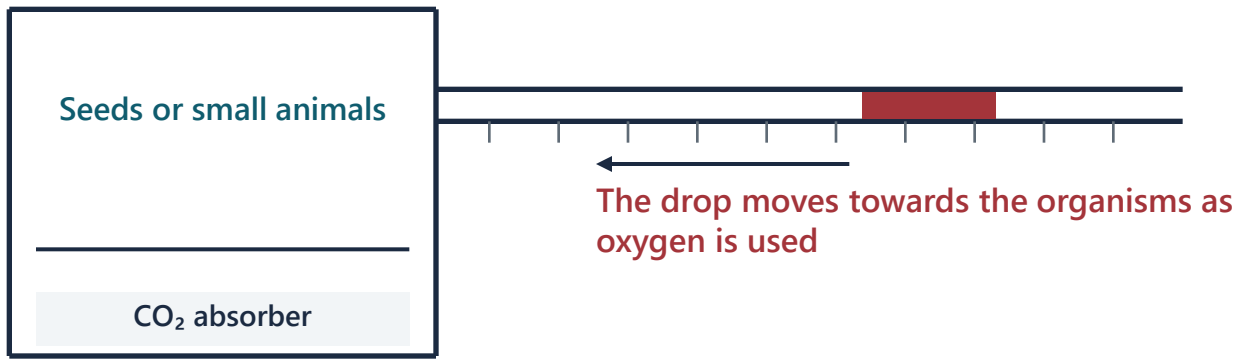
reload **myoglobin** and haemoglobin with oxygen.

Big question

This is why you keep breathing hard after a sprint: the anaerobic respiration used during the race left an oxygen debt that must be repaid. Fitter athletes repay it faster because their hearts and lungs deliver oxygen more quickly.

Measuring respiration with a respirometer

C1.2.6



Schematic, not to scale. Set up in a water bath.

Organisms are sealed in a tube with a chemical that **absorbs carbon dioxide** (such as soda lime).

As they respire, they take in oxygen, and the CO_2 they give out is absorbed, so the **volume of gas falls**.

This pulls a drop of coloured liquid along a scale. A **control** tube with glass beads instead of organisms shows any change caused by temperature or pressure.

Ethics

Animals such as woodlice must be kept away from the absorber, which is caustic, used only for a short time and returned to their habitat afterwards. Germinating seeds avoid most of these concerns.

Variables and calculating a rate

C1.2.6



Germinating seeds. Ramjchandran, CC BY-SA 4.0.

Variables you could investigate: **temperature**, type of organism, stage of germination, or the **mass** of organisms.

Variables to control: temperature (water bath), mass of organisms, volume of the tube and amount of absorber, time.

Worked example

A drop moves 45 mm in 15 minutes in a capillary tube with a cross-sectional area of 1 mm^2. Volume of oxygen used = 45 × 1 = 45 mm^3. Rate = 45 / 15 = 3 $mm^3\ min^{-1}$. Dividing by the mass of seeds (5 g) gives 0.6 $mm^3\ min^{-1}\ g^{-1}$, so tubes with different masses can be compared.

Working with data: peas at three temperatures

C1.2.6

Distance moved by the drop in a respirometer containing 10 g of germinating peas. Tube cross-section 1 mm^2. These data are invented for practice; the pattern is realistic.

Time / min	0	5	10	15
10 °C	0	6	12	18
20 °C	0	13	25	38
30 °C	0	24	49	73

1. Calculate the rate of oxygen uptake at 20 °C in $mm^3\ min^{-1}\ g^{-1}$.
2. Describe the effect of temperature on the rate.
3. Explain this effect.
4. Predict, with a reason, the result at 60 °C.

Check-in: types of respiration and measuring it

C1.2.5–6

1. Write the word equations for anaerobic and aerobic respiration in humans.
2. State three differences between anaerobic and aerobic respiration.
3. Explain why a sprinter keeps breathing hard after finishing a race.
4. State the role of the carbon dioxide absorber in a respirometer.
5. Explain why a control tube is used in a respirometer experiment.

How to use this

Write full sentences, then mark with a partner.

Common mistakes to avoid

Not this	But this
Respiration is breathing.	Respiration is a set of reactions in cells; breathing is ventilation.
ATP is a store of energy.	ATP is a short-term carrier; it is made and used within seconds.
Respiration makes energy.	Energy cannot be made; respiration releases it from carbon compounds to make ATP.
Anaerobic respiration in humans makes carbon dioxide.	In humans it makes lactate only.
Plants do not respire.	All living cells respire, including plant cells, day and night.

Check that you can...

1. **Explain** why ATP suits its role as the energy currency of the cell
2. **List** processes that use ATP
3. **Describe** the energy changes when ATP is hydrolysed and rebuilt
4. **Distinguish** between cell respiration and gas exchange
5. **Write** word equations for anaerobic and aerobic respiration in humans
6. **Compare** anaerobic and aerobic respiration
7. **Explain** how a respirometer works, including ethical issues
8. **Calculate** a rate of respiration from raw data, with units

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Write the word equation for anaerobic respiration in humans.
2. State two differences between anaerobic and aerobic respiration.
3. State the role of the CO_2 absorber in a respirometer.

Image credits

Mitochondria, mammalian lung – TEM.jpg – Louisa Howard – Public domain

ATP structure.svg – Mysid – Public domain

Animal mitochondrion diagram en.svg – Mariana Ruiz Villarreal LadyofHats – Public domain

Germinating seeds.jpg – Ramjchandran – CC BY-SA 4.0

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